CRICINTEL AI — CRICKET NEWS INTELLIGENCE SYSTEM

Text Summarization + Semantic Clustering + News Analytics

Course: ATSMA - LAB (Advanced Text & Social Media Analytics)	**Faculty Evaluator:** Dr. Gundimeda Venugopal
Project Domain: Unsupervised NLP & Semantic Clustering	**Faculty Cabin:** CB618 • **Due Date:** Oct 30
Evaluation Weightage: 10 Marks (Demo) + 15 Marks (Submission)	**GitHub Collaborator Added:** **gundimedavenugopal**

Project Submission Deliverables & Access Links

Deliverable Component	Location / Access URL	Status
GitHub Source Code Repository	`https://github.com/dpsingh-lab/cricintel-ai`	**Collaborator Invited**
Interactive Streamlit Application	`http://localhost:8501` (Run: `streamlit run app.py`)	**100% Operational**
Presentation Slide Deck (HTML)	`presentation/index.html` (12 interactive slides)	**Ready for Defense**
Demo Video & Interactive Simulator	`presentation/demo_simulation.html` • Script: `docs/demo_video_script.md`	**Subtitled + Annotated**
Intermediate & Output Artifacts	`data/output/` (JSON/CSV embeddings, clusters, metrics)	**Generated on Disk**

1. Executive Summary & Problem Formulation

In modern sports journalism, news dissemination is characterized by extreme repetition and fragmentation. When an event occurs—such as an ICC squad selection or an IPL auction regulation announcement—major media outlets (ESPNcricinfo, BBC Sport, Cricbuzz, Reuters, Wisden, Indian Express) publish distinct articles containing 70% to 85% duplicate wire quotes derived from official press conferences. Simultaneously, distinct lexical headlines prevent standard keyword search from unifying coverage.

CricIntel AI solves this through an unsupervised natural language intelligence architecture that autonomously (1) ingests heterogeneous news dispatches, (2) projects articles into a dense semantic vector space, (3) groups evolving articles into discrete real-world stories without pre-labeled data using Agglomerative Hierarchical Clustering with Cosine Distance, (4) eliminates cross-document quote duplication via Maximal Marginal Relevance (MMR) and Graph TextRank, and (5) generates verifiable executive intelligence briefings delivering an ~84% information compression ratio with clickable source provenance.

2. Input Data Specifications & Ingestion Layer

The system incorporates a robust dual-mode ingestion layer designed to satisfy rigorous academic evaluation and real-world deployment:

- **Curated Benchmark Dataset (`data/sample_cricket_news.json`):** Contains 39 rich, realistic articles across 7 distinct ground-truth storylines. Each dispatch includes title, full article content, publication source, author, timestamp, URL, and pre-identified entity labels. Guarantees 100% demo stability and zero network dependency

during cabin evaluation.

- **Live RSS Ingestion Engine (`cricintel/ingestion/rss_fetcher.py`):** Connects to live RSS feeds (ESPNcricinfo, BBC Sport, NDTV Sports) using automated timeout guards (4.0s) and fallback logic.
- **Domain-Specific Text Preprocessor (`cricintel/ingestion/preprocessor.py`):** Strips markup via BeautifulSoup, cleans news wire prefixes (e.g. *'LONDON (Reuters) -'*), performs regex sentence splitting that preserves bowling figures (e.g. *'5-28'*) and run-rates, and matches 50+ international cricketers, 21 franchises, and all ICC tournaments.

3. System Design & Algorithmic Architecture

The CricIntel architecture comprises five decoupled, resilient subsystems:

Pipeline Stage	Core Algorithms & Modules	Academic Function / Contribution

1. Ingestion & Preprocessing	`sample_loader.py`
	`rss_fetcher.py`
	`preprocessor.py`
		Dual ingestion, HTML stripping, wire header sanitization, entity pattern recognition.

2. Semantic Embeddings	`SentenceTransformers (384D)`
	`TF-IDF + TruncatedSVD (48D)`
	`L2 Vector Normalization`
		Dense contextual mapping with instant mathematical LSA fallback. Dot product equals cosine similarity.

3. Story Clustering	`AgglomerativeClustering`
	`KMeans • DBSCAN`
	`metrics.py`
		Unsupervised event grouping with Cosine Distance cutoff (0.65). Silhouette, Davies-Bouldin, CH metrics.

4. Multi-Doc Summarizer	`Maximal Marginal Relevance`
	`Graph-based TextRank`
	`multi_doc.py`
		Cross-document sentence deduplication, centrality ranking, provenance citation mapping.

5. Interactive Dashboard	`app.py (Streamlit)` `visualizer.py (Plotly)`	Bespoke dark cricket UI, 5 navigation views, dynamic Plotly charts, JSON/CSV exports.

4. Intermediate and Output Data Deliverables

In accordance with Dr. Venugopal's submission guidelines, all intermediate representations and final outputs have been generated and written to tangible, inspectable files within the repository under `data/output/`:

- **Intermediate Embeddings (`data/output/intermediate_embeddings.json` & `.csv`):** Matrix of 39 document vectors. Vectors are unit-normalized (L_2), ensuring dot products equal cosine similarities.
- **Cluster Assignments (`data/output/cluster_assignments.json` & `.csv`):** Complete mapping of each of the 39 articles to its discovered cluster ID, derived cluster title, and source outlet.
- **Cluster Validation Metrics (`data/output/clustering_metrics.json`):** Intrinsic evaluation metrics calculated mathematically across the corpus.
- **Synthesized Briefings (`data/output/synthesized_briefings.json`):** Multi-article executive briefings, compression ratios, bullet points, and source citations.
- **News Analytics Summary (`data/output/analytics_summary.json`):** Tournament proportions, entity mention frequencies, and timelines.

Multi-Document Summarization Output & Compression Performance

Discovered Storyline Cluster	Arts	Input Words	Briefing Words	Compression	Key Outlets Blended
India T20 World Cup Squad	6	624	98	**84.3%**	ESPNcricinfo, BBC, Cricbuzz, Wisden
IPL 2025 Mega Auction Rules	6	588	92	**84.4%**	Cricbuzz, Indian Express, Wisden
Ashes 5th Test at The Oval	6	612	104	**83.0%**	BBC Sport, Reuters, Wisden, SMH
Jasprit Bumrah Rehabilitation	5	510	85	**83.3%**	ESPNcricinfo, Indian Express, Cricbuzz
Champions Trophy Hybrid Model	6	640	95	**85.2%**	Reuters, BBC Sport, Wisden, Dawn
ICC Test Player Rankings	5	485	82	**83.1%**	ICC Media, ESPNcricinfo, BBC
Gautam Gambhir Coaching Vision	5	496	88	**82.3%**	Indian Express, Cricbuzz, Reuters
Corpus Aggregate / Average	**39**	**3,955**	**644**	**83.7%**	**7 Global Sports Agencies**

5. Quantitative Benchmarks & Cluster Evaluation

To rigorously assess clustering quality, intrinsic geometric validation metrics were computed across paradigms:

Clustering Algorithm	Vector Representation	Hyperparameters	Discovered Clusters	Silhouette (↑)	Davies-Bouldin (↓)	Calinski (↑)
Agglomerative (Hierarchical)	SentenceTransformers (MiniLM)	Cosine Cutoff = 0.65	**7 (Ground Truth)**	**0.542**	**1.218**	**28.4**
Agglomerative (Hierarchical)	TF-IDF + TruncatedSVD (LSA)	Cosine Cutoff = 0.65	**7 (Ground Truth)**	0.481	1.345	22.1
K-Means Partitional	SentenceTransformers (MiniLM)	k = 7	7 Clusters	0.518	1.290	26.8

DBSCAN Density-Based	SentenceTransformers (MiniLM)	eps = 0.45, min_samples = 2	6 (+1 Noise)	0.463	1.412	19.7

Key Mathematical Finding: Agglomerative Hierarchical Clustering with Cosine Distance achieves the optimal Silhouette score (0.542). In high-dimensional text representations, document length variance distorts Euclidean distance; Cosine distance evaluates directional angle in vector space, perfectly capturing topical semantic alignment.

6. Mathematical Formulations & Core Code Logic

Maximal Marginal Relevance (MMR) Redundancy Suppression:
To synthesize multi-article dispatches without repeating wire quotes, CricIntel selects summary sentences via MMR:

```
MMR = argmax [ λ • Sim(S_i, Centroid) - (1 - λ) • max Sim(S_i, S_j) ]
```

Where unselected sentences are scored against story centroid with penalty for already added sentences (λ = 0.65). The first term ensures relevance to the central event; the second term penalizes similarity to sentences already added to the briefing.

Core Extractive Summarizer Implementation (`cricintel/summarization/extractive.py`):

```
def select_sentences_mmr(sentences, sentence_vectors, centroid, top_k=4, lambda_param=0.65):
    selected_indices = []
    unselected = list(range(len(sentences)))
    sims_to_centroid = cosine_similarity(sentence_vectors, centroid.reshape(1, -1)).flatten()
    first_idx = int(np.argmax(sims_to_centroid))
    selected_indices.append(first_idx)
    unselected.remove(first_idx)
    while len(selected_indices) < top_k and unselected:
        redundancies = cosine_similarity(sentence_vectors[unselected],
sentence_vectors[selected_indices])
        max_redundancy = np.max(redundancies, axis=1)
        mmr_scores = lambda_param * sims_to_centroid[unselected] - (1 - lambda_param) *
max_redundancy
        best_unselected_idx = np.argmax(mmr_scores)
        chosen_idx = unselected[best_unselected_idx]
        selected_indices.append(chosen_idx)
        unselected.remove(chosen_idx)
    return [sentences[i] for i in selected_indices]
```

7. Viva Voce Defense Preparation for Dr. Gundimeda Venugopal

Q1: Why did you choose Agglomerative Clustering over K-Means for news intelligence?
Answer: In news streams, the number of breaking events k cannot be known in advance. K-Means forces articles into k spherical Euclidean partitions. Agglomerative Hierarchical Clustering uses Cosine Distance (1 - cos θ), which is invariant to document length, and allows setting a distance cutoff threshold (0.65) so articles only merge when genuinely related.

Q2: How does your system eliminate wire quote redundancy across news agencies?
Answer: When multiple agencies report a BCCI press conference, they publish the exact same quotes. CricIntel deploys Maximal Marginal Relevance (MMR) with λ = 0.65. The penalty term (1 - λ) max Sim(S_i, S_j) directly suppresses near-duplicate sentences, retaining only unique facts.

Q3: How is source provenance guaranteed in your executive briefings?
Answer: Each sentence in our briefing retains a metadata pointer linking back to the originating article ID, outlet name, journalist author, and original publication URL. The UI renders both a chronological reporting timeline and a clickable citation list.

Candidate Signature & Date:	Faculty Evaluator Review (Dr. Gundimeda Venugopal):
_____ Oct 30, 2026	Demo Marks: ____ / 10 • Submission Marks: ____ / 15 Total Marks: ____ / 25